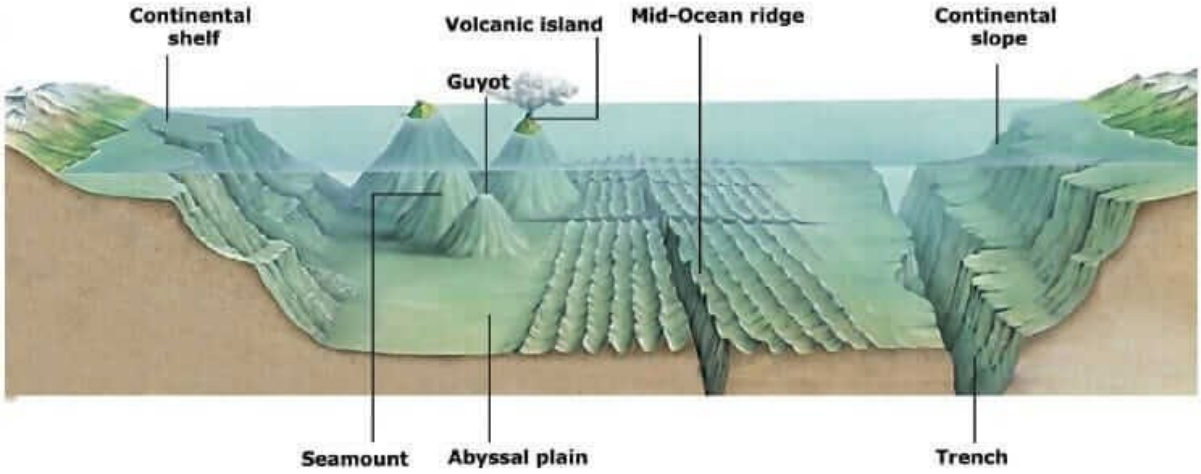
Table 12.2 : Climatic Types According to Koeppen

Group	*Type*	*Letter Code*	*Characteristics*
A-Tropical Humid Climate	Tropical wet	Af	No dry season
	Tropical monsoon	Am	Monsoonal, short dry season
	Tropical wet and dry	Aw	Winter dry season
B-Dry Climate	Subtropical steppe	BSh	Low-latitude semi arid or dry
	Subtropical desert	BWh	Low-latitude arid or dry
	Mid-latitude steppe	BSk	Mid-latitude semi arid or dry
	Mid-latitude desert	BWk	Mid-latitude arid or dry
C-Warm temperate (Mid-latitude) Climates	Humid subtropical	Cfa	No dry season, warm summer
	Mediterranean	Cs	Dry hot summer
	Marine west coast	Cfb	No dry season, warm and cool summer
D-Cold Snow-forest Climates	Humid continental	Df	No dry season, severe winter
	Subarctic	Dw	Winter dry and very severe
E-Cold Climates	Tundra	ET	No true summer
	Polar ice cap	EF	Perennial ice
H-Highland	Highland	H	Highland with snow cover

Oceanography (6:03 PM)

- Ocean bottom Topography
- Ocean temperature
- Salinity
- Ocean Deposits
- Coral reefs
- Oceanic Currents and Circulations
- El Nino and related phenomena
- Tides

Ocean Bottom Topography (6:24 PM)



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- **<u>Continental shelf</u>**
- (a) These are submerged shallow portions of the continental crust with depths between 150-200 m
- (b) It is rich in sunlight, oxygen, and nutrients, which favours the growth of plankton.

- (c) The width of the continental shelf is dependent upon the slope of the adjacent land.
- (d) It is wider along the coastal plain, and narrow along the mountainous coast.
- (e) The continental shelf is rich in biodiversity and is known for its rich fishing grounds.
- **Continental Slope**
- (a) It is the boundary between continents and oceans, formed by the outer edge of the continents.
- (b) They are much steeper than the continental shelf
- **Continental Rise**
- (a) It is the base of the continental slope, with a lower slope.
- (b) It continues into abyssal plains.

Submarine Canyon (6:54 PM)

- These are the canyons, formed along the continental shelf and slope, because of the turbidity currents of the river, entering the ocean.
- **Abyssal Plain**
- They are also called Oceanic basins
- They are tectonically inactive oceanic floors, with relatively smooth surfaces.
- The average depth varies between 3000-6000m
- **Seamounts**
- These are isolated volcanic features and volcanic mountains formed due to volcanic activities, along hotspots and subduction zones.
- **Volcanic Islands, Island Arc, Archipelago**
- These are formed due to volcanic activities along the subduction zones and hotspots.
- (* Formation has been covered in Plate tectonics)
- **Guyots**
- While seamounts are sharply pointed, guyots are flat-topped hills, with summits below the water level.
- **Mid-Oceanic Ridges**
- They are formed along Ocean-Ocean divergent plate boundaries.
- **Trenches**
- They are formed along the Ocean-Ocean convergence and Ocean-continental convergence plate boundaries.

Temperature (7:17 PM)

- **Differences concerning the Atmosphere**
- (a) The water heats up slowly and cools down slowly.
- (b) The sunlight penetrates only upto 200m
- (c) The albedo of the open ocean is low compared to land.
- (d) The temperature distribution through mixing is slower.
- **Factors**
- **(a) Latitudes**
- In Lower latitudes, higher insolation results in higher temperatures.
- Along, higher latitudes, lower insolation, and lower intensity cause lower temperatures.
- **(b) Winds**
- Winds help in the distribution of temperature across the globe.

- For Example, in the regions where westerlies are active, water temperature remains higher, than in the region of polar easterlies.
- **(c) Ocean currents**
- They help in heat balance, between the equatorial and polar regions.
- Warm ocean currents increase the temperature, and cold Ocean current decreases it.
- **(d) Role of Landmass**
- The seas which are surrounded will have higher water temperatures, than open ocean waters.
- **<u>Vertical distribution</u>**

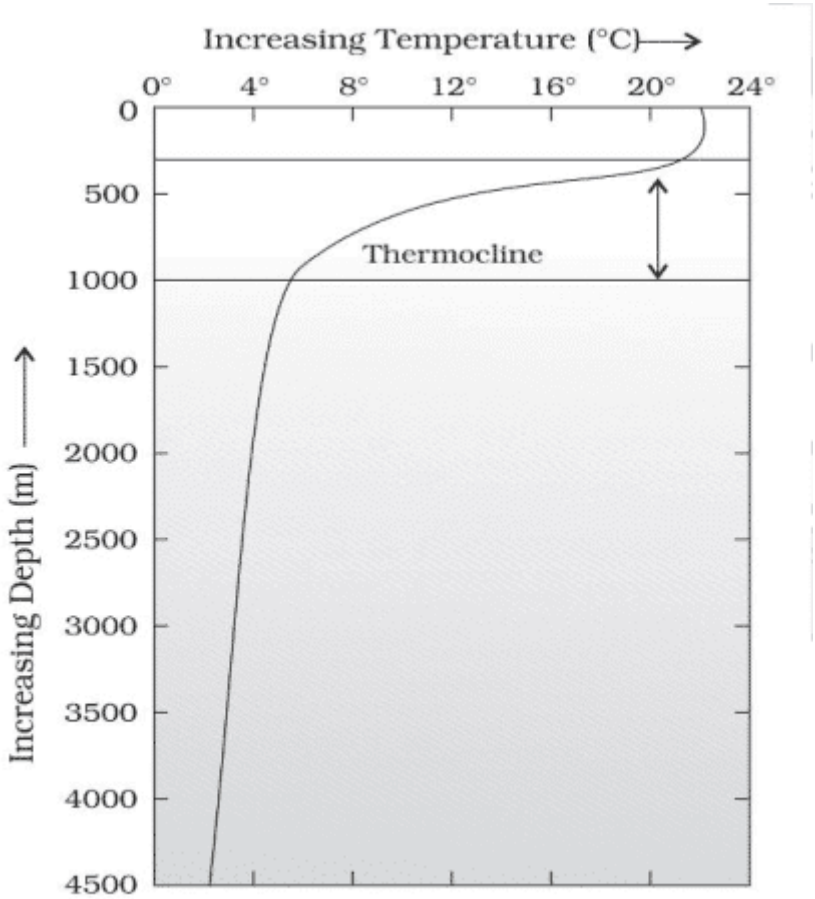


Figure 13.3 : Thermocline

-
- The first layer is called Epilimnion (up to 200m)
- The second layer is called Thermocline (or Metalimnion) (200-1000m)
- The third layer is called Hypolimnion (beyond 1000m)
- *Epilimnion:*
- It extends up to 200m in depth.
- The average temperature is between 20-25 degrees celsius.
- It is a zone of constant sunlight and mixing due to winds.
- It is a zone of uniform temperature.
- It is the zone of Life
- *Thermocline:*
- It is the twilight zone, because of the very low amount of sunlight.
- The temperature decreases rapidly with depth.
- *Hypolimnion:*

- It is a zone of uniform and very cold temperatures.
- The average temperature is nearly 4 degrees Celsius
- It is because seawater attains maximum density at 4 degrees Celsius.
- It is also known as a dark zone.
- **Horizontal Distribution**
- The surface temperature lies between 20-25 degrees Celsius in tropical regions and decreases gradually towards the polar region.
- The three distinctive layers are visible along tropical regions.
- However, along polar regions, there will be a single uniform layer of cold temperature.

Salinity (7:50 PM)

- Salinity is the number of grams of dissolved salts in a thousand grams of seawater.
- It is measured as parts per thousand (ppt).
- The average salinity of Earth's ocean is 35 ppt.
- **Sources of salinity**
- Sources of addition
- (a) Sediments brought by rivers, which get dissolved over a long period of life.
- (b) Volcanic eruptions along the oceanic floor
- (c) Evaporation in Ocean
- The salinity of oceans always remains constant.
- The factors increasing salinity are balanced by the factors decreasing salinity.
- Sources reducing salinity
- (a) Addition of river water
- (b) Rainfall
- (c) Ice melting
- (d) precipitation of salts along the oceanic floors.
- Dissolved salts in Sea Water (Decreasing Order): Chlorine, sodium, sulfate, magnesium, Calcium

The Topic for the Next class: Oceanography (continued)

Geography Class 31

Revision of the previous class (05:11 PM)

Factors influencing salinity (05:23 PM)

- a)Temperature: The higher temperature will result in Higher salinity
- b)Wind speed: Higher wind speed causes a higher rate of evaporation resulting in higher salinity.
- c)Ocean Currents: Help distribute salinity from one region to another.
- **Regions for Higher salinity**
- Also, Refer World map as drawn in the class.
- **Sub-Tropical Regions**: Due to the Higher rate of Evaporation. Examples: Mediterranean sea, Red sea, Persian Gulf, and Sub-tropical regions of the Atlantic and Pacific oceans.
- **Regions with below-average salinity:** North sea, Baltic sea, Berring sea, Arctic Ocean, Southern Ocean
- **Regions with Exceptionally High salinity:** Lake van, Dead sea, and Great salt lake.

Oceanic Deposits (05:37 PM)

- **Terrigenous Deposit:** Deposits are derived from the land and found in oceans which is why also known as Lithigeneous. Example: Sand, Gravel, Mud, silt.

- **Biogeneous Deposits**: Shells, Corals, Ooze(These are the deposits derived from calcareous and silicious shells and skeletal remains of marine microorganisms. Biogeneous deposits are also called organic deposits.
- **Hydrogenous Deposits:** Example: Salt, Polymetallic nodules.
- **Cosmogenous Deposits:** Meteorites debris.

Coral Reefs (05:47 PM)

- Coral Reefs are masses of limestone and dolomite accumulated by a lime-secreting organism called coral polyps.
- They are developed due to the symbiotic relationship between coral polyps and zooxanthellae algae.
- The coral reef provides habitat to nearly 31 phyla of Animals leading to Rich biodiversity, Therefore they are also called **rainforests of oceans.**
- **Conditions**
- a)Temperature: 20 degrees celsius, Not beyond 30 degrees North and south latitude.
- b)Depth: Maximum depth between 200-250 feet below sea level.
- c)Sediment-free water: Therefore coral reefs are not found near deltas.
- d)Salinity: Average of 35 parts per thousand.
- e)A platform on which the corals can build reefs i.e Continental shelves.
- Corals are not found in the western margins of the continents due to cold ocean currents.

Distribution (06:15 PM)

- Atlantic Ocean: Around the Caribbean sea, Mexico, and the surrounding regions.
- Indian Ocean: Red sea, Persian gulf, western and Northern Indian ocean, A&N islands, Lakshwadeep, Gulf of Mannar, Gulf of Kutchh, Maharashtra Coast, and Goa Karnataka coast.
- Southeast Asia and Pacific: Covering Coral Triangle from Indonesia, Malaysia, Philippines, and Australia(Great Barrier Reef), The islands of the pacific ocean.
- **Types of Coral Reefs (06:24 PM)**
- **Fringing Reefs**
-

- A narrow and discontinuous patch of coral reef located near the coast. Example coral reef of the Caribbean sea.
- **Barrier Reef**
-
- A wide and continuous coral reef located far from the coast and is separated by a lagoon, Example: Great Barrier Reef.
- **Atoll Reef**
-
- A circular Coral reef developed around a sea mount or Gyot or island. Ex: Lakshwadeep, Maldives, and islands of the pacific ocean.

Coral Bleaching (06:34 PM)

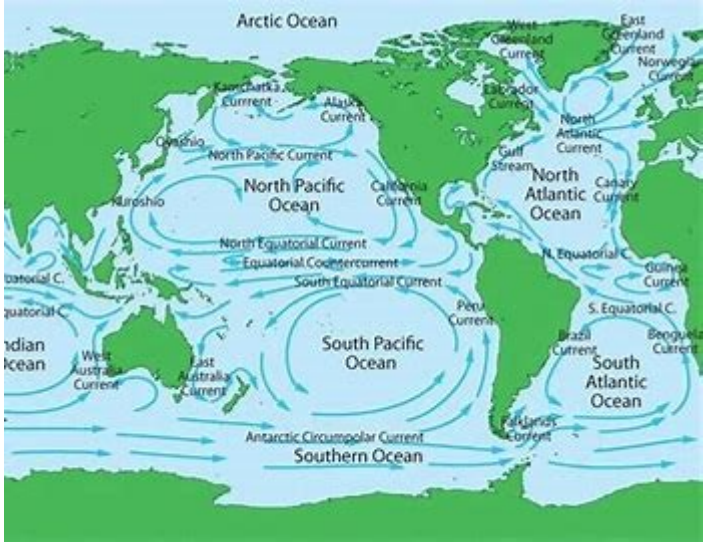
- Corals receive their colouration from zooxanthellae.
- Coral Bleaching refers to the loss of colour of corals due to the stress-induced expulsion of symbiotic algae.
- The reasons behind coral bleaching are:
- Changes in water temperature, salinity, Increased sedimentation, Increased pollution, and Ocean acidification.Increased incidence of cyclones, Storms, and Elnino.
- Examples of Coral Bleaching: Between 2014 to 2017, Due to El Nino the Great Barrier Reef was bleached by nearly 50%.
- In 1997-98 the Northern Indian ocean lost the majority of coral reefs.

Oceanic circulation (07:18 PM)

- **Surface/Ocean Current**
- An ocean current is the general movement of the ocean's surface water in a definite direction over long distances.
- Types of ocean currents:
- a)Warm Ocean current: Moves from Equator to pole
- b)Cold ocean current: Moves from poles to the Equator.
- **Factors affecting ocean currents**
- Winds: The steady blowing of winds drag the surface water in its direction and brings about the surface flow and sets the water currents in motion.
- The shape of the Coastline and Topography: Modify the ocean currents' direction.

- Differences in temperature, salinity, and density: Cause the water to move and reduce variations such differences guide the movement of ocean currents.
- Coriolis force: It causes ocean currents to deflect right in the Northern hemisphere and left in the southern hemisphere.

Ocean Currents of the world (07:45 PM)



- **Currents of the Atlantic Ocean**
- Under Trade winds influence
- a)North Equatorial current
- b) South Equatorial current
- c)Counter Equatorial current which goes along the Equator in the west-to-east direction.
- d)Florida current: Anteles +Gulf current, Along the coast of Florida.
- e)Gulf stream: Towards the Northeast direction, USA coast.
- f)North Atlantic Drift: Gulf stream takes a total right turn once it crosses Newfoundland under the influence of westerlies and Coriolis force.
- g)Norweigian current: Along the coast of Norway
- h)The canary current: A cold current, Along the canary islands, flowing southwards.
- Two cold ocean currents Labrador and the East Greenland current. Labrador joint Gulf stream and East Greenland joins North Atlantic Drift.
- Seaweed in the Gyre grows which is known as sargasso that is what the sea is known as Sargasso sea.

The Topic for the next class: El Nino

Geography Class 32

Brief Discussion on the Previous Class (5:13 PM)

Currents of the South Atlantic Ocean (5:22 PM)

- South equatorial current, which moves along Brazil forms the **Brazilian current**. It is a warm current.
- The Brazilian current gets deflected towards the left in the southern hemisphere (Due to Coriolis force) and joins the **West Wind Drift**.
- West Wind Drift is a cold current, which circulates the continent of Antarctica. (It flows under the influence of Westerlies).
- The Brazilian current once when comes under the influence of West wind drift is called **South Atlantic Drift.**

- Near the coast of Africa, it turns upward and is known as **Cold Benguela Current.**
- Benguela current is responsible for the formation of the Namib and Kalahari Deserts.
- Cold Benguela current joins the South Equatorial current.
- Falkland Current flows along the eastern coast of the South American continent.
- Gyre is formed in the South Atlantic Ocean,

Currents of the Pacific Ocean (5:34 PM)

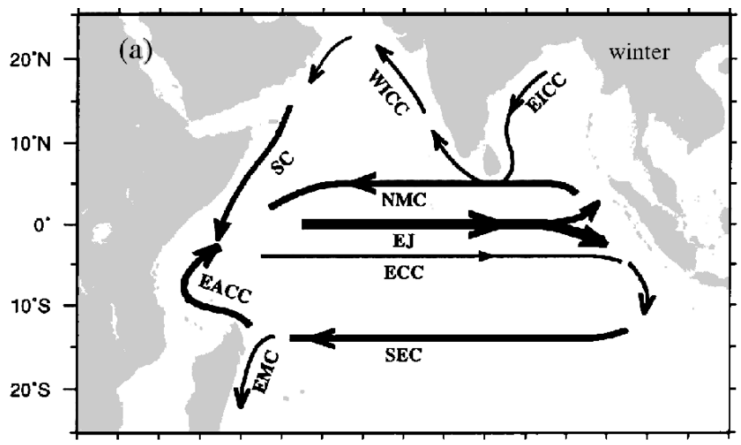
- North Equatorial current gets divided by the archipelago and turns rightward. This current is known as Warm **Kuroshio Current** (along the coast of Japan).
- It is a warm current moving along the coast of the Philippines and Japan.
- The **North Pacific Current** is an ocean current that flows west-to-east between 30 and 50 degrees north in the Pacific Ocean.
- It gets divided near the North American Continent.
- One branch moves upward and is known as **Warm Alaskan Current**.
- Another branch moves southward and is known as the cold **California Current.**
- Cold California Current joins the North Equatorial Current. therefore a gyre is formed here.
- Near Sakhalin Island, there are two cold currents- **Oyashio Current** and **Okhotsk Current**.

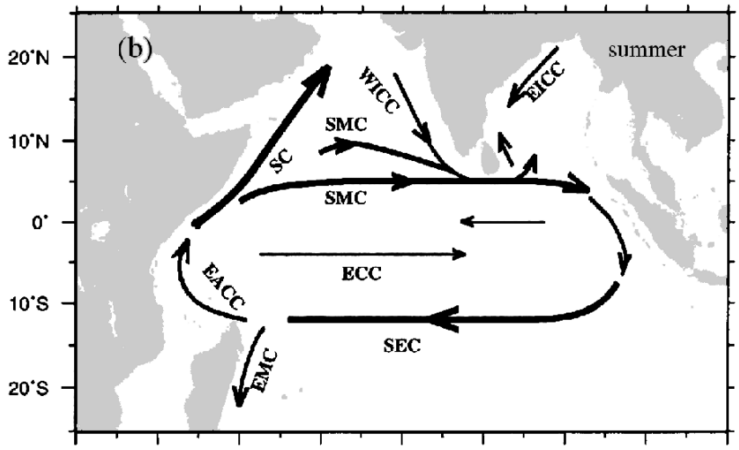
Southern Pacific Ocean

- The southern equatorial current moves down and is called Warm **East Australian Current.**
- Coming down, it joins West wind drift and is known as Cold **South Pacific Drift.**
- Then it flows along the coast of the South American continent and is known as the cold **Humboldt Current.**
- Then, finally, it joins the South Equatorial current.

Indian Ocean currents (5:48 PM)

- **Southern Indian Ocean**
- There is a Southern equatorial current.
- It gets divided into two currents by Madagascar: Mozambique current and Madagascar Current
- Mozambique and Madagascar combine to form the Agulhas current.
- It flows along with West Wind Drift and then, moves along the western coast of Australia. It is then known as Western Australian Current.
- Western Australian Current then, merges with the Southern Equatorial current.
- **Northern Indian Ocean**





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- Here, the wind reversal takes place. Also, there is huge dominance of land in the Northern Indian Ocean.
- **<u>Winters</u>**
- We have North Equatorial Current, South Equatorial current, and Counter equatorial current during the winter.
- North equatorial current flows westward, above the equator.
- Counter equatorial current flows in between the North and South Equatorial Currents.
- The Northeast monsoon drives the water of the Bay of Bengal and Arabian sea to circulate in an anti-clockwise direction as the Northeast monsoon drift.
- **<u>Summers</u>**
- The entire water of the Northern Indian Ocean comes under the influence of the Southwest Monsoon.
- This produces a clockwise flow of the southwest monsoon drift.
- During this time, there are no north equatorial currents and counter equatorial currents.

Significance of Ocean Currents (6:25 PM)

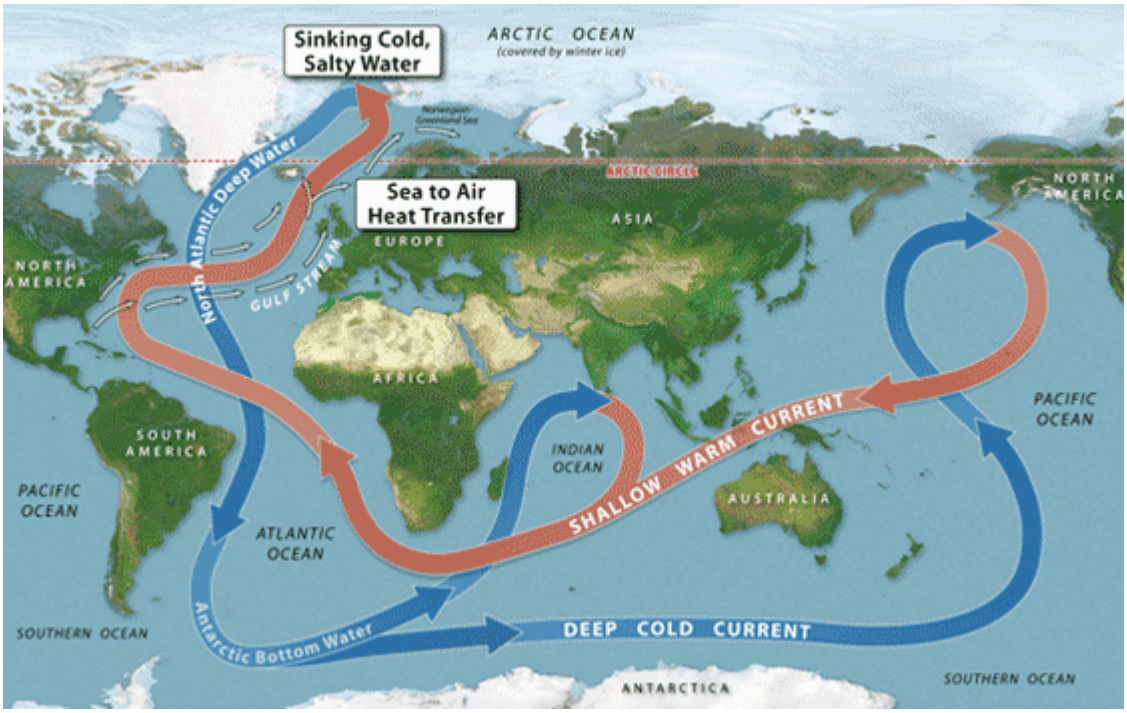
- (a) The continuous movement of water from east to west in the tropical region causes cold water upwelling, along the eastern parts of oceans.
- This causes a drop in the temperature
- (b) Accumulation of warm waters along the western parts of the ocean in tropical regions, leads to a rise in water temperature.
- (c) The western parts of the Ocean within the tropical regions, experience favourable conditions for the formation of cyclones.

- (d) The cold ocean currents along the western coast in sub-tropical regions, suppress precipitation through the desiccation effect.
- It contributes to the formation of deserts. For example, the Kalahari desert and Namib desert due to the Benguela current, the Atacama desert due to the Humboldt current, etc.
- (e) Warm and Cold ocean currents merge near the temperate regions to create favourable conditions for the development of large-scale fishing banks.
- For Example Grand Bank (near Newfoundland Island due to the merging of labrador current with the gulf stream)
- (f) The off-shore winds along the western margins of the continent, in tropical and sub-tropical regions push the surface water away.
- This causes the cold water from the bottom to rise upwards to the surface.
- This brings cold and nutrient-rich water to the surface through upwelling.
- The reasons for upwelling support large-scale fishing growth. For example, the Humboldt current along the coast of Peru and chile.
- (g) Ocean Currents help in the global distribution of temperature and salinity.
- (h) the warm ocean currents in the temperate region along the western coast of continents, result in mild winters and keep the port ice-free.
- For example, North Atlantic Drift along Britain.
- (i) Ocean currents can be used in navigation, to find directions in the open oceans.
- (j) The merging of Warm and Cold Ocean currents results in fog formation, which may obstruct ocean navigation.

Upwelling and downwelling (7:17 PM)

- **<u>Downwelling</u>**
- The on-shore winds pile up the water near the coast, causing the sinking of water below the surface layer.
- Downwelling is more active where warm and cold ocean currents meet.
- **<u>Upwelling</u>**
- The off-shore winds push the surface water away and cause the cold water from the bottom to rise from the surface.
- Upwelling is more active in the regions of cold ocean currents.
- In Polar regions, ice formation leaves the water saltier and denser causing it to sink.

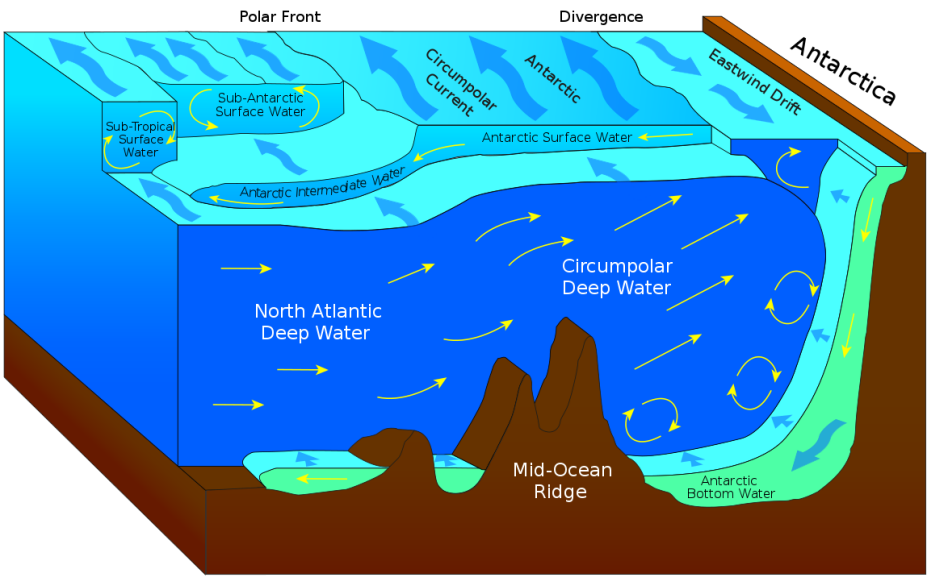
Thermohaline Circulation (7:21 PM)



- The deep ocean currents are driven by differences in the water density, which is controlled by the temperature and salinity and is called thermohaline circulation.
- The circulation begins near polar regions, of the downwelling where ice formation leaves the water saltier and denser leading to the sinking.
- This underwater current moves in a single and continuous belt across the different oceans.
- They are connected to surface water currents in the regions of upwelling.
- They are also called **Great Ocean Conveyor belts.**

Atlantic Meridional Overturning Circulation (7:46 PM)

- AMOC is a part of the thermohaline circulation, in the Atlantic ocean.
- The recent report of the IPCC, talks about the weakening of AMOC, due to global warming.
- **Factors responsible for the weakening**
- (a) The temperature difference between the tropical and polar regions is decreasing due to the warming of the arctic region.
- This weakens the north Atlantic drift.
- (b) The excessive addition of freshwater due to increased melting of ice in polar regions.
- This reduces downwelling as the freshwater is less dense.
- **Impacts**
- (a) Western European countries will experience harsh winters and frozen ports.
- (b) Western parts of Oceans in the tropical regions, will experience a rise in sea levels.
- (c) Increased occurrence of cyclones in tropical waters.
- (d) Disturbance of fishing zones.
- (e) Disturbance of marine ecosystem.
- **Water Mass (7:53 PM)**



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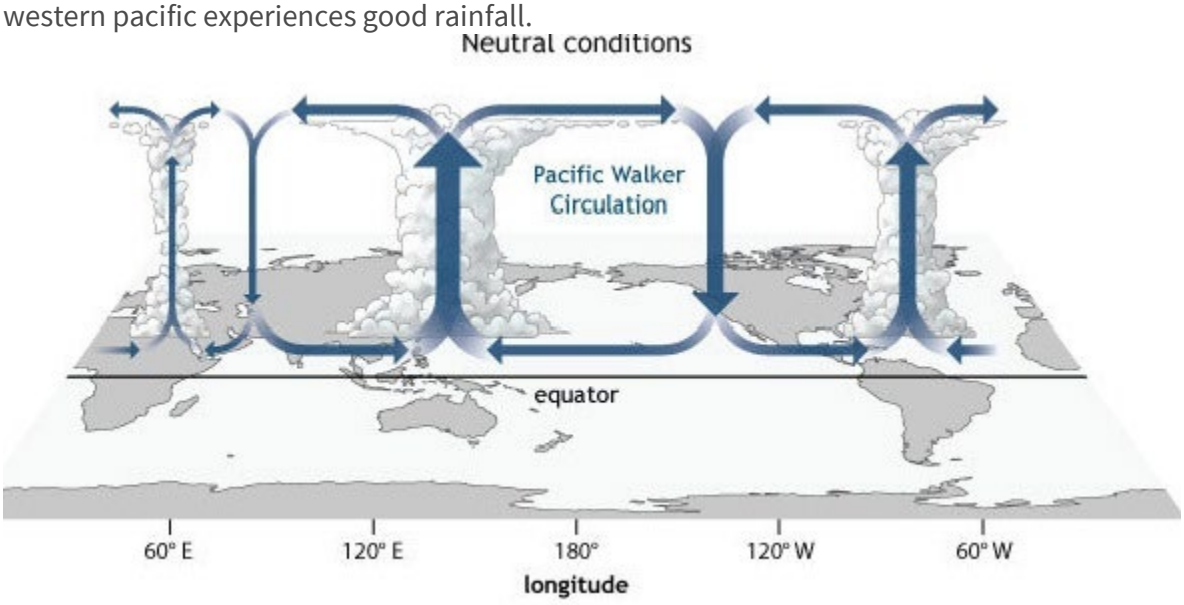
- It is a body of water with uniform characteristics of temperature and salinity.
- They are three types:
- (a) surface water mass- For example, Antarctic Surface Water mass
- (b) Intermediate water mass: For example Sub tropical Intermediate water mass of the Atlantic Ocean.
- (c) Deepwater mass: For example, North Atlantic Deep Water Mass
- water mass helps in the generation and maintenance of thermohaline circulation.
- they result in upwelling and downwelling.
- The movement of water mass helps distribute temperature, salinity, and oxygen.
- The uniformity of conditions, within water mass, supports plankton and coral growth.

The Topic for the next Class: El Nino and the related phenomena
Geography Class 33

Discussions on the last Class (5:10 PM)
Walker circulation (5:39 PM)

- Along the western coast of South America, the ITCZ is usually above the equator throughout the year, this results in strong trade winds.
- The offshore trade winds push the water, away from the coast, causing cold water upwelling, along the eastern pacific in addition to existing cold ocean currents.
- This reduces the water temperature and the air above this cold water is dry and stabilized, and Convection is suppressed.
- Water flows westward as southeast trade winds blow toward the western pacific region.
- This water gets heated up and rises through convection causing cloud formation and precipitation along the western pacific.
- The rising air flows eastward, in the upper troposphere, and descends along the eastern pacific completing the cell.
- This results in high pressure and low temperature, in the eastern pacific ocean.
- And high temperature and low pressure in the western pacific region.
- This results in the convective cell, known as walker circulation.

- In normal conditions, the eastern pacific coast experiences, drought, and the western pacific experiences good rainfall.



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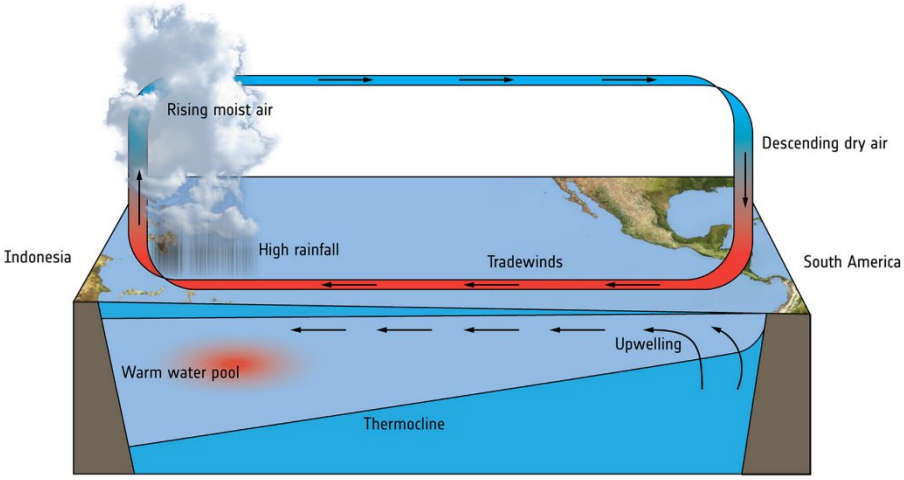
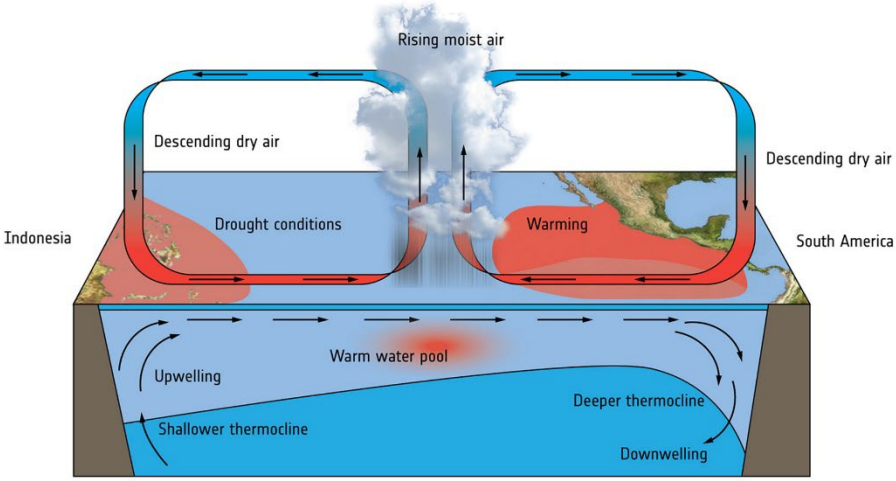
El Nino (6:07 PM)

- Once in 3-7 years in October-November, the ITCZ moves too much south to the equator.
- This results in the weakening of trade winds and disruption of upwelling.
- The warm tropical surface waters, earlier flowing towards the west, flow back eastwards, causing a further reduction in the upwelling.
- This results in warm surface water on the coast, of central and South America, which is called as El-Nino.
- The appearance of El Nino causes the air pressure to drop over the eastern pacific coast and rise over the western pacific.
- This see-saw variation, of air pressure, is called El Nino southern oscillation (ENSO)
- El Nino causes heavy precipitation along the eastern pacific and drought conditions along the western pacific.



Normal conditions



El Niño conditions

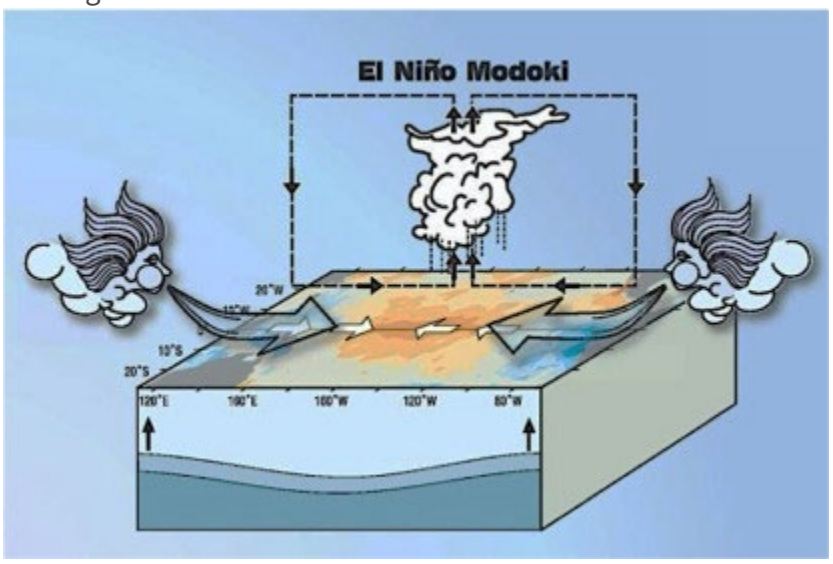
-
- After one or two years the enhanced Hadley circulation strengthens the trade winds and increases the upwelling, bringing back the normal condition
- **<u>Impacts of El NINO</u>**
- (a) The disturbance of the walker circulation causes the disturbance in global weather patterns and the convective cycles are also disturbed in other oceans.
- (b) It causes drought and dry conditions in India and Australia.
- (c) It negatively impacts Indian Monsoon, causing delays in monsoon, a decrease in precipitation as well as an increase in the number of breaks in monsoon.
- (d) It causes heavy precipitation along the coastal deserts of Peru and Chile.
- (e) The dry condition in India and Australia triggers forest fires and bushfires.
- (f) the warm water along the eastern pacific along Peru and chile coast inhibits the growth of plankton and algae and suppresses upwelling devastating the marine ecosystem, and killing fish on large scale.
- (g) It results in a catastrophic impact on the fishing and fertilizer industry.
- (h) the decomposition of fish releases hydrogen sulfide, causing dark waters and making it toxic.
- (i) El Nino years cause mass coral bleaching due to disturbance of the stable conditions and higher temperature.

- (j) It results in severe dust storms in India and Australia.
- (k) During El NINO, high-pressure conditions in the Caribbean Sea, Western Pacific, and Bay of Bengal suppress the cyclone formation.
- The strongest recorded El Nino was in 1982-83, 1997-98, and 2014-16, 2018-19.

La Nina (6:27 PM)

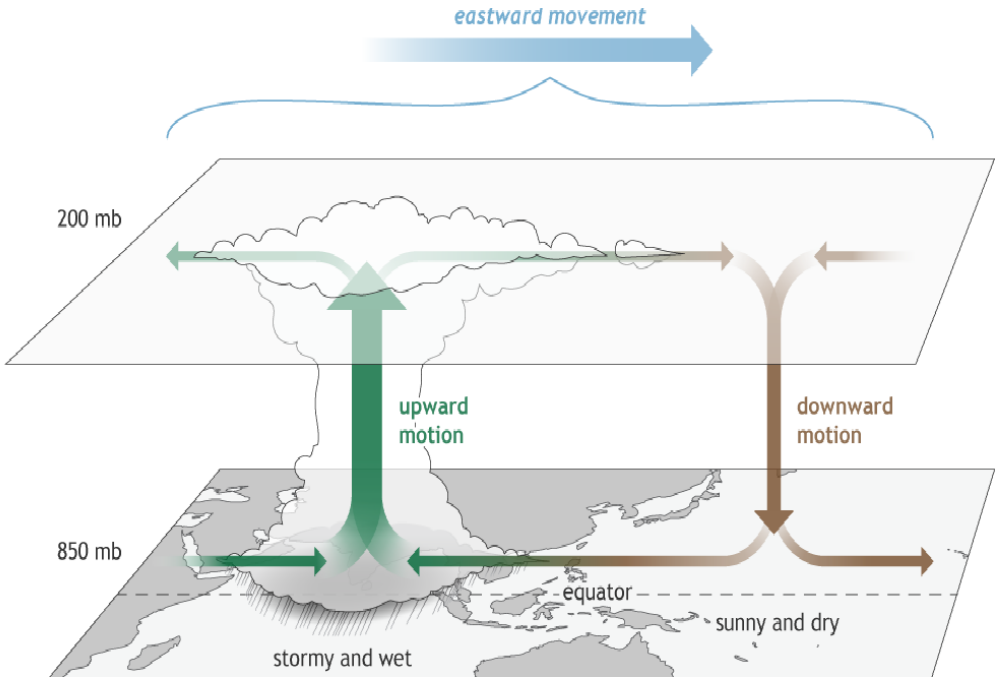
- El Nino is usually followed by La Nina.
- It is the intensified high pressure, and cold temperature along the eastern pacific and lower low pressure and higher high temperature along the western pacific.
- This causes very heavy precipitation, in India and Australia, resulting in floods.
- The La Nina Years, 2020-21, 21-22, 22-23. This is called Triple Dip. (La Nina consecutively in three years)

El Nino Modoki (7:21 PM)

- Modoki means similar but different.
- While El Nino is characterized by strong warming in the eastern pacific ocean, El Nino Modoki results in warming in the central pacific with cooling in the eastern and western pacific oceans.
- It creates a two-cell Walker circulation over the tropical pacific region, with precipitation in the central part and dry conditions in the eastern and western parts.
- While El Nino results in diminished hurricanes in the Caribbean sea, El Nino Modoki results in increased hurricanes.
- During this time, the number of cyclones is more in the Arabian sea, than in the Bay of Bengal
- 

Madden Julian oscillation (MJO) (7:39 PM)

- Unlike ENSO which is stationary, MJO is an eastward moving disturbance of clouds, rainfall, winds, and pressure that traverses the planet in the tropics and returns to its initial starting point in 30-60 days on average.
- MJO has two parts, the convective part with enhanced rainfall and the subsiding part with suppressed rainfall.
- MJO can influence the timing and strength of monsoons, as well as tropical cyclones.
- MJO can increase or decrease the impact of El Nino on India.

- 

Indian Ocean Dipole (IOD) (7:44 PM)

- The difference in sea-surface temperature between the Western Arabian sea and the eastern Indian Ocean, near Indonesia and Australia is called as Indian Ocean Dipole.
- Similar to MJO, it is a coupled ocean and atmospheric phenomenon.
- **Positive IOD**
- Warm sea surface in the western part compared to the east.
- Less rainfall in Indonesia and Australia.
- It is good for Indian Monsoon.
- Reduces the impact of El Nino on India.
- Arabian sea experiences a higher number of cyclones than the Bay of Bengal.
- **Negative IOD**
- Warm sea surface along the eastern part than the western part.
- More rainfall in Indonesia and Australia.
- Negatively impacts the Indian Monsoon.
- Increases the effect of El Nino on India.

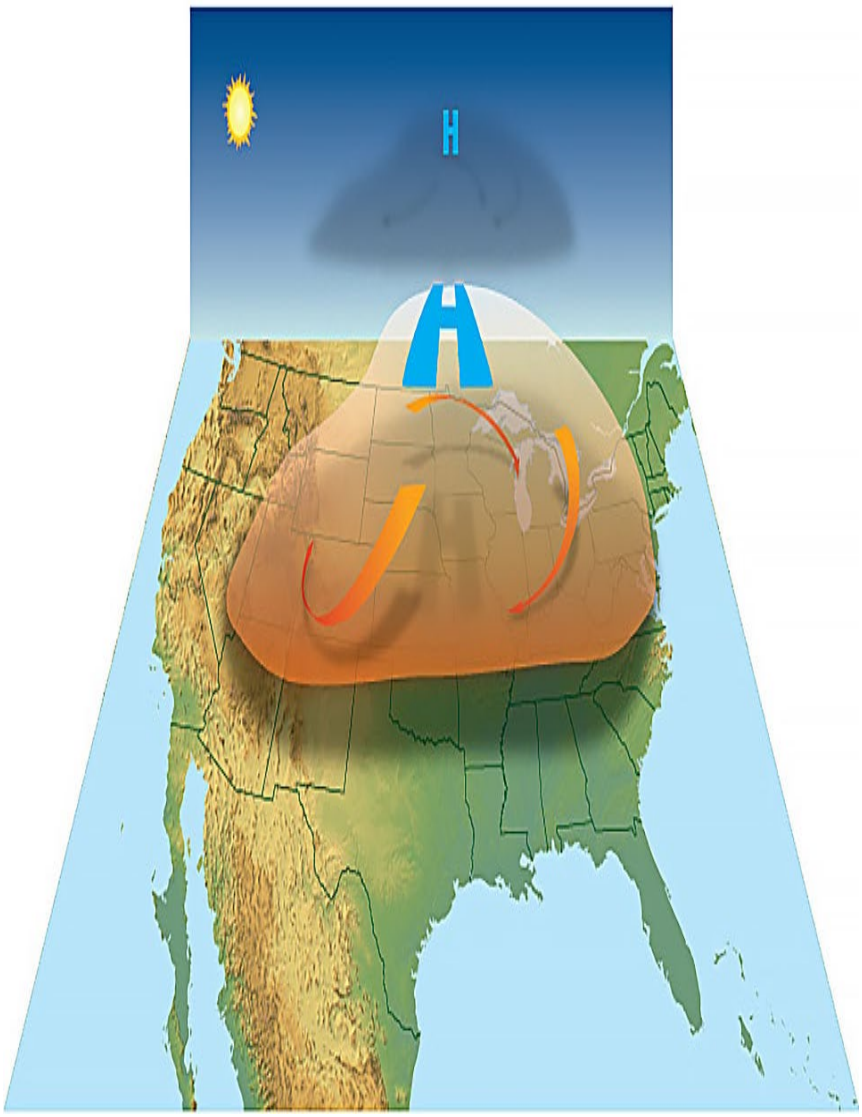
The topics for the Next Class: Heat Domes, Biogeography

Geography Class 34

Discussion on the last class (5:12 PM)

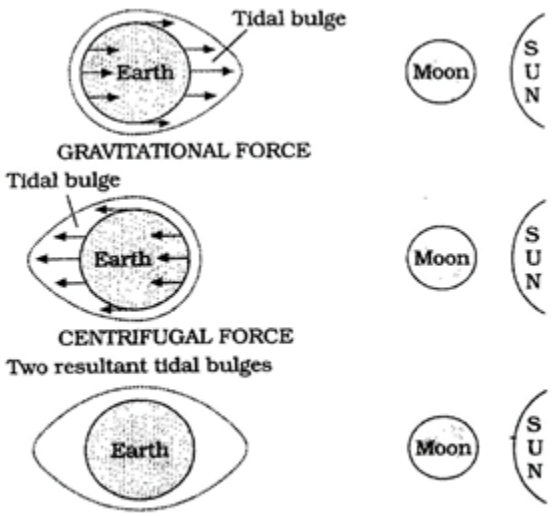
Heat Dome (5:36 PM)

- The phenomena of heat dome that occurred in Canada and the USA in 2021.
- It occurs when the atmosphere traps hot air like a lid
- It is due to when the combined effect of La Nina and weakened Jet streams.
- La Nina results in strong subsidence of hot air, along the eastern pacific.
- Jet streams when weakened start to meander in their path.
- This weakened Jet stream, meandering too much allows the hot air from the tropical region to reach the higher latitude and traps the hot air descending due to La Nina too.

- This results in higher than normal temperatures causing the heat dome.
- 

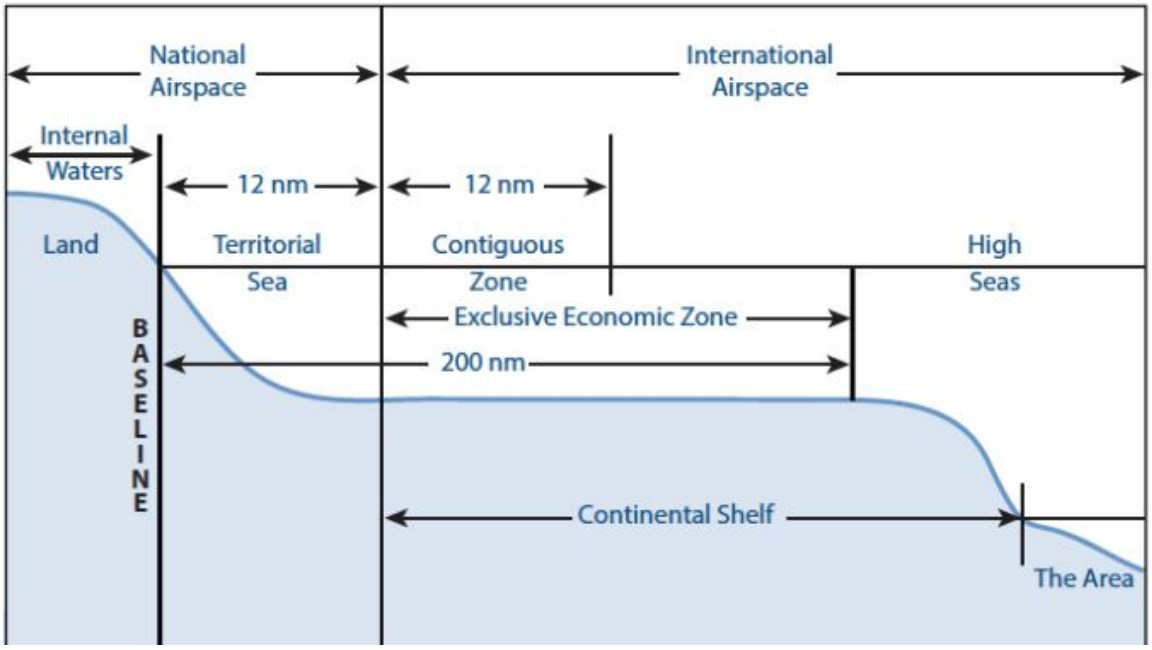
Tides (6:03 PM)

- The regular rise and fall of water levels in the world's oceans resulting from gravitational attraction by the moon and the sun is called Tides.
- The combined gravitational pull of the Sun and Moon pulls the oceanic water in a bulge.
- The centrifugal force due to the earth's revolution pulls another bulge, in the opposite direction.
- The bulges in one direction create low water in another.
- As the earth rotates, the position of high and low waters change across the surface.



- Gravitational and Centrifugal Forces
- **<u>Factors Affecting Tides</u>**
- (a) Gravitation pull of Sun and Moon
- (b) Relative position of the Sun and Moon
- (c) Centrifugal force of earth's revolution.
- (d) Uneven distribution of Water
- (e) Ocean Bottom topography
- **<u>Types of Tides</u>**
- (a) <u>Spring tide</u>
- These are very high, high tide, and very low tides.
- They are experienced when the sun, earth, and moon are in a straight position.
- This is called syzygy
- The conjunction is when the earth, moon, and sun are in E-M-S
- Opposition is when alignment is M-E-S
- (b) <u>Neap tide</u>
- These are smaller high tides and smaller low tides.
- A neap tide is when the sun and moon are in a perpendicular direction.
- The position is called Quadrature.
- The tides which are observed twice a day are called **semi-diurnal tides**
- Those which are observed once a day are called **Diurnal tides.**
- **<u>Significance of Tides</u>**
- (a) Tidal power generation
- (b) It helps in navigation for the tidal ports.
- (c) It helps in fishing activities.
- (d) Helps in maintaining higher biodiversity along the tidal range.
- (e) Helps in the natural cleansing of the coasts.
- (f) It restricts the delta formation, as it doesn't allow the sediments to accumulate.
- (g) At other places it causes coastal erosion.

Maritime Zones (6:42 PM)

- 

- Baseline
- It is the average line where the land ends and the ocean begins.
- Internal Waters
- It includes all the water bodies along the landward side of the baseline.
- Territorial Sea
- It is 12 nm from the baseline.
- A country exercises sovereign jurisdiction on the bottom, surface, and air. However, innocent passage is allowed with restrictions.
- Contiguous Zone
- It is up to 24 nm from the baseline.
- Sovereign jurisdiction over the bottom and surface but not in the air.
- Exclusive Economic Zone
- up to 200 nm from the baseline.
- A country can exploit the resources on the ocean floor.
- It extends beyond 200 nm if the continental shelf crosses it.
- It will be allowed till the edge of the continental shelf.
- High seas
- International waters come under International Seabed Authority.

Biogeography (7:12 PM)

- It is the study of the soil in relation to vegetation.
- Soil
- The loose surface material consists of inorganic and organic matter and is the source of water and nutrients for the vegetation.
- True soil is characterized by distinct horizons and the capacity to support vegetation.
- Horizon
- Horizon is a distinguishable layer in the soil that has certain chemicals and a distinct colour.
- Regolith
- The layer of unconsolidated materials is derived from weathering of the parent rock.